



Ethical Analysis of Annotated Social Media for Human Rights

Major Group – 51

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Problem Statement

Detecting and categorizing human rights violations from vast, noisy, multilingual, and unstructured social media data poses critical challenges in ethical annotation, interpretability, and adversarial robustness, since existing manual solutions are not fulfilling and deficient, calling for a solution of automation is needed.

Objectives

To build a responsible system for monitoring human rights violations by:

- Developing a Multilingual Dataset to Identify Human Rights Violation.
- Evaluate Large Language Models and Experts for Annotation Process.
- Enhance capability and reasoning of LLMs.
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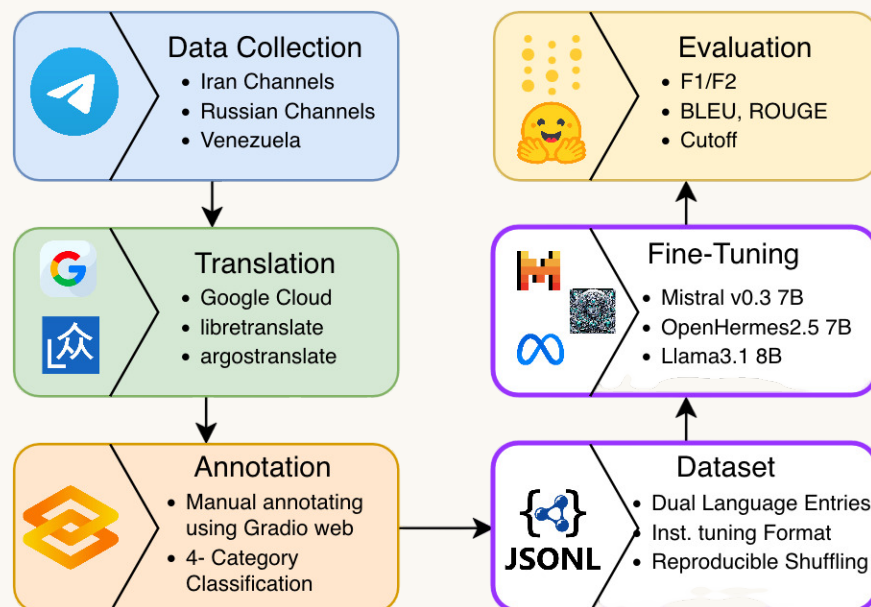


Fig. 1 Proposed Methodology

Methodology

Multilingual posts are fetched from Telegram using API, translated using Google Translation API and annotated using an Interface. The annotations are chosen on dependency of event. These are then finetuned on basis of classification criteria using 8B/7B LLM models with reasoning enhancement.

Implementation Details

The system detects HRVs form Telegram Posts. Steps introduce translation of posts, annotation on Gradio and finetuning on LLMs.

Three LLMs fine-tuned: Mistral-7B-Instruct, LLaMA-3.1-8B-Instruct, and OpenHermes-2.5-7B. The training used LoRA for PEFT. Classifying posts into HRV categories such as civilian injury, destruction of civilian objects, torture/execution, and mistreatment of detainees

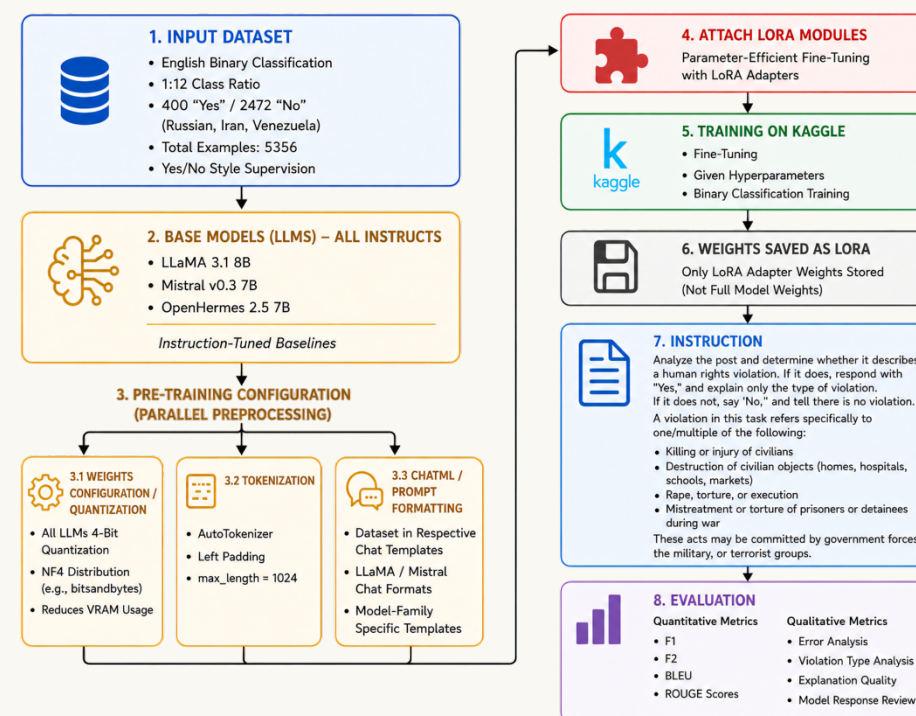


Fig. 2 Training Architecture

Results

Model	ACC.	F1	F2	BLEU
Mistral v0.3	0.92	0.90	0.88	0.56
Llama3.1	0.79	0.77	0.74	0.75
OpenHermes2.5	0.94	0.93	0.92	0.86

Conclusion and Future Scope

This project presents an AI-based framework for detecting possible human rights violations from social media text. It reduces the burden of manual monitoring and can support researchers. The use of open-source LLMs and efficient fine-tuning makes the system practical even with limited computing resources. It can be improved by increasing dataset size, adding more languages and integrating multimodal analysis for images and videos, confidence scoring, better explainability and real-time monitoring.

Reference

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- [2] A. Vaswani *et al.*, "Attention Is All You Need," 2023.
- [3] J. Wei *et al.*, "Finetuned Language Models Are Zero-Shot Learners," Feb. 2022, [Online]. Available: <http://arxiv.org/abs/2109.01652>

Publications (In Submission)

A. Singh, S. Srivastava, D. Dhiman, A. Dogra, and R. Narwal, "Fine-Tuned Open LLMs in Human Rights Violation Detection" submitted to the Proceedings of the AAAI Conference on Artificial Intelligence (AAAI-27), Jan. 2027